

# **A Graph Neural Surrogate for Longitudinal Train Dynamics**

Toward fuel-optimal control of freight consists

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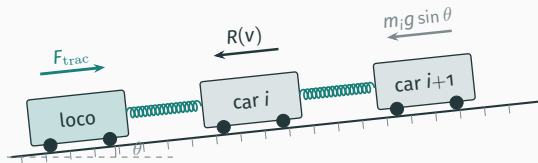
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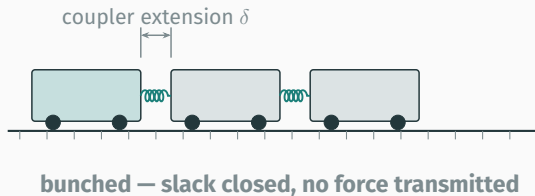
# A freight train is not a rigid body

- Tens to hundreds of coupled masses, each sampling its **own** grade and curvature.
- Couplers carry **slack** — free travel before any force is transmitted.
- What the operator cares about: **fuel, arrival time, peak coupler force.**

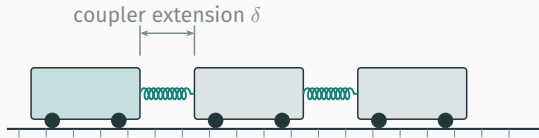


$$m_i \dot{v}_i = F_{\text{ext},i} + F_{i-1} - F_i$$

# Slack is the whole problem



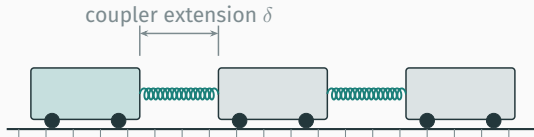
# Slack is the whole problem



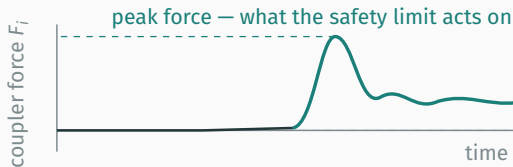
**slack taken up – couplers reach the deadband edge**



# Slack is the whole problem



**run-out — force spikes the instant slack closes**



A run-in/run-out event is what breaks a knuckle — and it is a **derivative discontinuity** that wrecks adaptive solvers.